CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EC)

November 2013

EC 403 Mobile & Satellite Communication

Date: 16.11.2013, Saturday | **Time: 10.00 a.m. To 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of a scientific calculator is allowed.

SECTION - I

Q - 1 (a) Define any seven following terms– [07]

(1) Base station (2) Setup channel (3) Forward channel (4) Reverse channel (5) Roamer (6) Half Duplex system (7) Full Duplex system (8) Mobile switching center (9) Page

Q - 2 (a) What are channel assignment strategies in cellular system? Classify and explain the various channel assignment strategies in detail. [07]

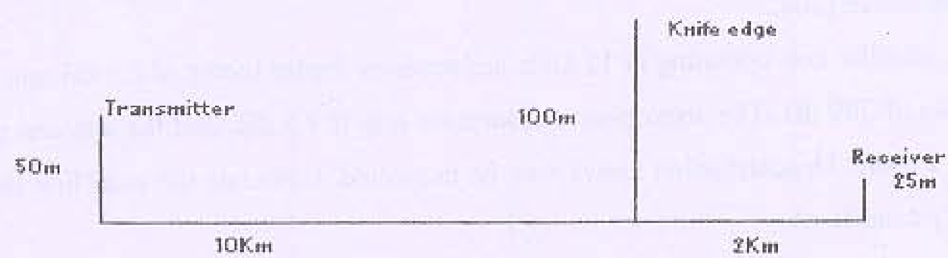
(b) Discuss any two points in brief for improving coverage and capacity in cellular system. [07]

OR

Q - 2 (a) For given path loss exponent (1) n=4 and (2) n=3, find the frequency reuse factor and the cluster size that should be used for maximum capacity. The signal to interference ratio of 15db is minimum required for satisfactory forward channel performance of cellular system. There are six co-channel cells in the first tier, and all of them are at the same distance from mobile. Use suitable approximations. [07]

(b) What is interference and system capacity in cellular system? Explain adjacent channel interference in brief. [07]

Q - 3 (a) Given the following geometry, determine (1) diffraction parameter (v) due to knife edge diffraction and (2) the height of obstacle required to induce 6db diffraction loss. Assume f=900MHz. [07]



(b) How reflection affect by dielectric and perfect conductor in mobile radio propagation? [07]

OR

Q - 3 (a) Explain Fresnel zone geometry and knife edge diffraction model with neat diagram. [07]

(b) Discuss various factors influencing small scale fading in mobile radio propagation. [07]

SECTION – II

Q - 4 (a) Explain Following Terms of Global System for Mobile Communication:(1) Home Location Register (2) Visitor Location Register [04]

(b) Define Frequency Division Multiple Access (FDMA) System. Determine the Features of FDMA system. [03]

Q - 5 (a) In GSM, each frame consists of 8 time slots, each time slot contains 4 start bits, 5 stop bits, 28 training bits and 18 guard bits and 4 burst of 58 bits. Data transmitted at 300 kbps. Find (1) Number of overhead bits per frame (2) Total number of bits per frame (3) Frame Rate (4) Time duration of Slot (5) Frame Efficiency. [07]

(b) Explain the "Location Updating" for Global System for Mobile Communication. Explain Diversity Feature of IS –95 Code Division Multiple Access Systems. [07]

OR

Q - 5 (a) Explain Offset Quadrature Phase Shift Keying (OQPSK) and Minimum Shift Keying (MSK) with necessary figures. [07]

(b) Draw the Block Diagram of a simplified communications system using an adaptive equalizer at receiver. [04]

(c) Define following terms:(1) Error Detection Codes (2) Error Correction Codes (3) Vocoders [03]

Q - 6 (a) Explain Different Types of Satellite Orbits with necessary figures. Enlist advantages of Satellite communication. [07]

(b) Define Following terms: (1) Geostationary Orbit (2) Apogee (3) Perigee (4) Sub Satellite Path. [04]

(c) Calculate the Gain in decibels of a 3-m paraboloidal antenna operating at a frequency of 10 GHz. Assume Aperture efficiency of 0.70 [03]

OR

Q - 6 (a) Explain Antenna Misalignment Loss, Feeder Loss and Free Space Transmission Loss in the Space Link. [07]

(b) A satellite link operating at 12 GHz has receiver feeder losses of 2.5 dB and a free-space loss of 209 dB. The atmospheric absorption loss is 1.5 dB, and the antenna pointing loss is 2.5 dB. Depolarization losses may be neglected. Calculate the total link loss for clear-sky conditions. [04]

(c) A satellite downlink at 10 GHz operates with a transmit power of 8 W and an antenna gain of 49.2 dB. Calculate the EIRP in dBW. [03]

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